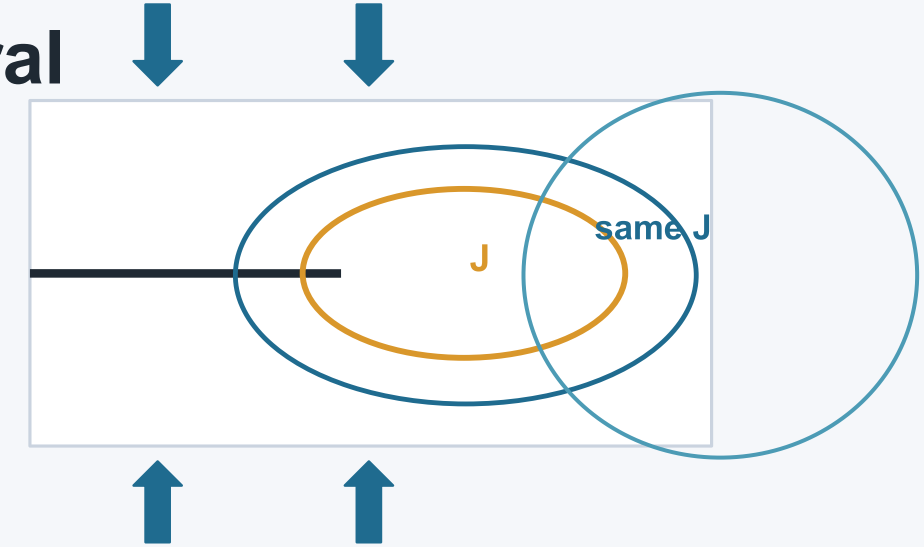
Rice 1968

A Path-Independent Integral

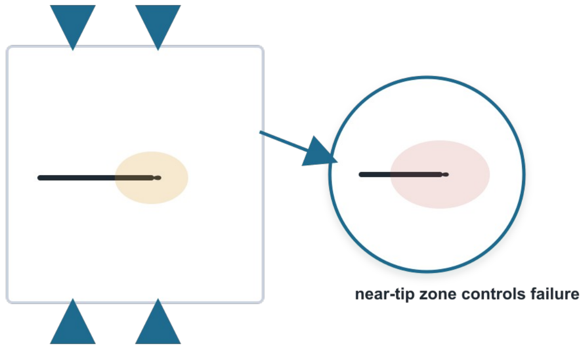
The J-integral and the birth of nonlinear fracture mechanics



Path-independent contour integral
Bridge from energy release to crack-tip severity
A practical language for elastic-plastic fracture

Why This Paper Matters

Failure begins in a tiny near-tip region
Loads and geometry are usually known far away
Plasticity makes the local field hard to solve



Rice connects the global problem to the local crack-tip danger.

The Problem Before the Paper

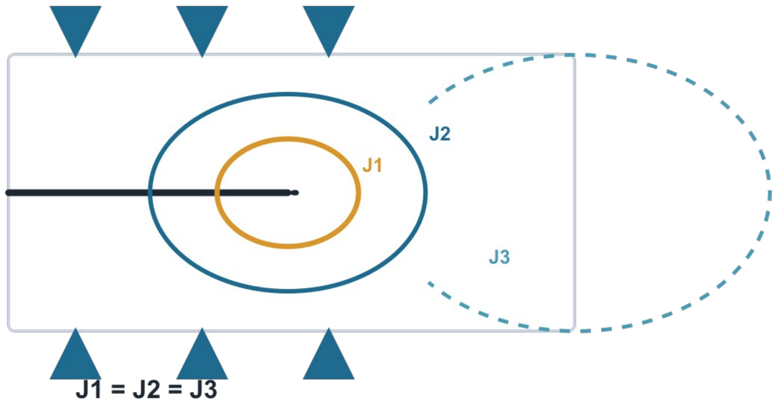
- Linear elastic fracture used K and energy release rate
- Elastic-plastic metals develop concentrated near-tip strain
- Full nonlinear boundary-value solutions are difficult

local stress and strain field is hard to solve directly

Can we characterize the tip without solving every local detail?

Rice's Core Move

Draw any valid contour around the tip
Integrate stored energy and boundary work
The value is the same: J1 = J2 = J3



Path independence is the central idea.

How to read J

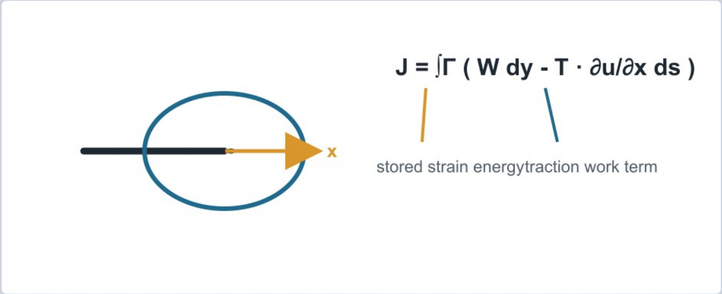
Intuitive Example

Evaluate where convenient. Interpret where failure begins.

The Essential Equation

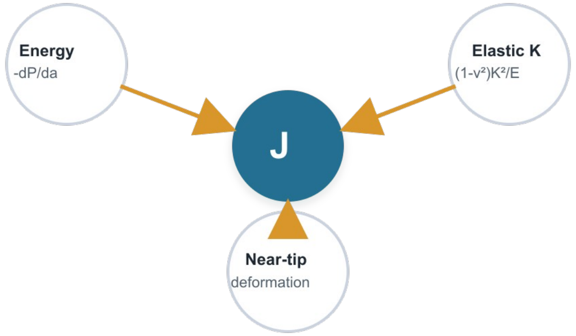
$$J = \int_\Gamma \left(W\,dy - T \cdot \partial u/\partial x \, ds \right)$$

W: strain energy density
T: traction on the contour
u: displacement field
x: crack-extension direction



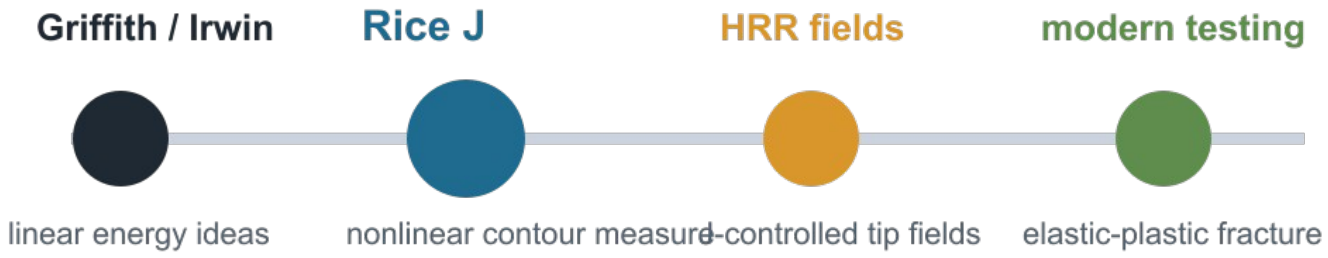
What J Connects

Energy rate: $J = -dP/da$
Small-scale yielding: $J = (1 - \nu^2)K^2/E$
Near-tip severity: averaged local deformation measure



Rice generalized classical fracture mechanics rather than replacing it.

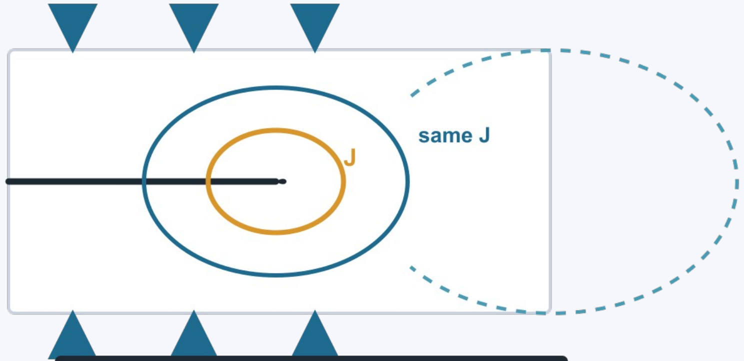
What Changed, and What It Does Not Solve



Assumptions: 2D fields, no body forces, traction-free crack faces, deformation-theory material behavior.

Final Takeaway

A hard local crack problem becomes a contour-energy measure
Evaluate J where it is convenient
Interpret J where failure begins



One contour integral. One crack-driving measure.

That quantity is J.